

Weather Shocks, Supply Dynamics, and Price Persistence: Evidence from U.S. Lettuce Markets

Yuhan Wang*

April 2026

Abstract

This paper constructs a weekly dataset that integrates USDA shipment data, FOB prices, and weather data to study weather-driven supply and price dynamics in U.S. lettuce markets from 2010 to 2026. Using an empirical strategy that combines market-level aggregation for prediction with district-level panel variation for identification, we document three results. First, simple linear models outperform tree-based machine learning methods. OLS achieves substantially lower prediction error than XGBoost and Random Forest (RMSE \$4.36 versus \$6.67 and \$6.81), reflecting strong price persistence and a data-generating process that is largely linear conditional on lagged prices. Second, weather effects on prices are present but attenuated in aggregate data. In district-level panels, frost events increase weekly prices by \$2.33 ($p < 0.001$), while these effects are diluted in market-level regressions due to price persistence and spatial aggregation. Third, these patterns are robust across lettuce varieties, including Romaine, and across alternative validation strategies. These results show that down-

*Department of Agricultural and Resource Economics, University of California, Davis. Email: yhn-wang@ucdavis.edu.

stream market outcomes, shipment volumes and FOB prices, can be used to identify weather-driven supply responses even in the absence of direct yield measures.

Contents

1	Introduction	5
2	Literature Review	6
3	Market Background	8
4	Data	10
4.1	Sources	10
4.2	Production Districts	11
4.3	Variable Construction	12
4.4	Summary Statistics	12
5	Methodology and Empirical Strategy	13
5.1	Dual-Layer Design	13
5.2	Market-Level Prediction Model	14
5.3	Change Model	15
5.4	District-Level Panel	15
5.5	Feature Construction	16
5.6	Validation Strategy	17
6	Results	17
6.1	Market-Level Price Prediction	17
6.2	Price Persistence and Hidden Weather Effects	19
6.3	Volume Prediction	21
7	Mechanism: Weather, Supply, and Price	21
7.1	District-Level Identification	22
7.2	Why Level Models Mask Weather Effects	23

8	Robustness	24
8.1	Romaine Lettuce	24
8.2	Expanding vs. Rolling Window	25
8.3	Coverage and Prediction Error	25
8.4	The 2022 Extreme Year	27
8.5	Feature Stability	27
8.6	Missing Price Prediction	28
9	Conclusion	29
	Appendix	34
A	Variable Definitions	35
B	Full OLS Regression Output	36
C	Out-of-Sample RMSE by Year	37

1 Introduction

Fresh produce supply chains are exposed to weather risk, but translating weather shocks into quantitative supply and price signals remains difficult. For most fresh vegetables, the data infrastructure that enables high-frequency crop monitoring in grain markets simply does not exist. There are no futures contracts, no weekly crop progress reports, and no county-level yield estimates. Buyers, distributors, and supply chain managers rely instead on trade reports, informal market intelligence, and experience. However, it remains unclear whether downstream market data can be used to recover supply responses to weather shocks.

This paper studies how weather shocks affect supply and prices in the U.S. iceberg lettuce market. Lettuce is the largest fresh vegetable crop in the United States by volume and is produced year-round through a seasonal rotation between California (spring through fall) and Arizona (winter). We construct a weekly dataset that integrates USDA truck shipment volumes, FOB shipping point prices, and gridded weather data from PRISM, covering six production districts over the period 2010–2026.

Our empirical strategy combines market-level aggregation for prediction with district-level panel variation for identification. At the market level, we aggregate all districts into a single weekly time series and estimate price and volume prediction models using OLS, XGBoost, and Random Forest. At the district level, we retain a three-district panel and use fixed effects to identify the causal effect of local weather shocks on prices. Both layers are necessary because aggregation improves data coverage (from 78% to 98% of weeks with observed prices) but dilutes the weather signal.

First, simple linear models outperform tree-based machine learning methods in level prediction tasks and perform comparably in change models. The best market-level price model achieves an out-of-sample RMSE of \$4.36, a 14.7% improvement over a naive autoregressive baseline. XGBoost and Random Forest perform substantially worse than the baseline in level models, and in change models all methods converge to similar performance. This reflects a data-generating process dominated by strong price persistence and approximate

linearity, under which additional model complexity does not improve predictive performance and reduces robustness in extreme periods.

Second, weather effects on prices are present but obscured by price persistence in aggregate data. In level models, the three price lags together account for nearly half of total feature importance, leaving weather variables with little apparent explanatory power. When we model week-to-week price changes instead, removing the autoregressive component by construction, weather variables become substantially more important. At the district level, a frost event increases the local weekly price by \$2.33 ($p < 0.001$), while this effect is attenuated in market-level aggregates. The mechanism runs from weather to local supply disruption to local price increases, which then persist through the autoregressive structure of the market.

Third, all results are robust across lettuce varieties. Replicating the analysis with Romeaine lettuce produces consistent coefficient signs and comparable, and in fact lower, prediction errors (RMSE of \$2.61 vs. \$4.36 for Iceberg).

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the institutional structure of the U.S. lettuce market. Section 4 presents the data sources and variable construction. Section 5 lays out the empirical strategy. Section 6 reports the main results. Section 7 discusses the mechanism linking weather, supply, and prices. Section 8 presents robustness checks, and Section 9 concludes.

2 Literature Review

This paper contributes to four related strands of literature: the econometrics of weather and agricultural production, the transmission of weather shocks to prices, climate impacts on specialty crops, and the use of machine learning in applied economics.

A large literature studies how weather affects agricultural production, primarily through crop yields. Schlenker and Roberts (2009) show that the relationship between temperature

and yields of major U.S. crops is highly nonlinear, with extreme heat driving large output losses. Subsequent work extends this framework using improved weather measures and broader datasets (e.g., Roberts, Schlenker and Eyer, 2013), and finds that weather variation explains a substantial share of yield fluctuations. More recent studies document that climate change has already reduced global agricultural productivity (e.g., Lobell, Schlenker and Costa-Roberts, 2011; Ortiz-Bobea et al., 2021). However, this approach relies critically on high-frequency yield data, which are typically available only for major row crops at annual frequency and are not available for most fresh vegetables such as lettuce.

A related literature examines how weather shocks translate into market outcomes such as prices. Early evidence is provided by Roll (1984), who show that futures prices respond to weather shocks. Roberts and Schlenker (2009) and Roberts and Schlenker (2013) estimate supply and demand elasticities using weather-driven yield shocks, finding that demand is highly inelastic and that small supply disruptions generate large price movements. More recent work shows that prices also respond to expectations about weather before harvest (Letta, Montalbano and Pierre, 2022). Most of this literature relies on yield shocks to identify price effects. In contrast, we show that weather effects on prices can be identified directly from market outcomes, and that these effects are attenuated in aggregate data due to price persistence.

Despite their economic importance and greater sensitivity to short-run weather variation, perishable specialty crops have received relatively little attention in this literature. Lobell, Cahill and Field (2007) estimate climate effects on yields of several California crops, including lettuce, using annual data, while Kistner et al. (2018) document that specialty crops are particularly sensitive to extreme weather events such as freezes and excess rainfall. These studies focus on annual yield responses. We extend this work by using weekly data to study short-run dynamics and by focusing on market outcomes, prices and shipment volumes, which provide a high-frequency proxy for supply in settings where yield data are unavailable.

Finally, a growing literature examines when machine learning methods improve upon tra-

ditional econometric approaches. Mullainathan and Spiess (2017) emphasize that machine learning excels in prediction tasks with complex, nonlinear relationships, while economic structure remains essential for interpretation. Storm, Baylis and Heckelei (2020) show that machine learning methods perform well in high-dimensional agricultural prediction settings, and Crane-Droesch (2018) find gains in crop yield prediction when nonlinearities are important. Our results provide a counterpoint. In the lettuce market, tree-based methods (XGBoost and Random Forest) underperform both OLS and a naive baseline. This reflects the fact that price dynamics are dominated by a strong autoregressive component that is approximately linear. Once persistence is removed by differencing, all methods perform similarly, indicating that the remaining weather-price relationship is also close to linear. More broadly, our findings contribute to understanding when additional model flexibility improves predictive performance and when simple linear models are sufficient.

3 Market Background

The U.S. lettuce market is highly geographically concentrated, with California and Arizona accounting for over 95% of domestic production (USDA National Agricultural Statistics Service, 2023). Production is organized around a seasonal geographic rotation between two primary regions.

Salinas-Watsonville, California is the dominant production district. Located in the Salinas Valley on the central California coast, it benefits from a mild maritime climate well suited to lettuce during the spring, summer, and fall months (roughly April through November, corresponding to weeks 15–50 of the year). It supplies the majority of national shipments during this period.

Western Arizona (Yuma) takes over during the winter months (roughly November through March, weeks 45–15). The region’s warm desert climate allows lettuce production when Salinas is too cold, and shipment volumes are comparable to Salinas during peak winter

weeks.

Imperial Valley, California serves as a supplementary desert production district, with a seasonal pattern similar to Yuma but smaller volumes. Three other California districts, Santa Maria, Oxnard, and the San Joaquin Valley, contribute smaller and more intermittent shipments.

The annual rotation between California and Arizona is largely calendar-driven and follows a stable pattern over time. Production transitions occur around weeks 15–20 (as Salinas ramps up and Arizona winds down) and weeks 45–50 (as Salinas winds down and Arizona ramps up). These transition periods are associated with temporary supply contractions and elevated price volatility. They are also particularly sensitive to weather shocks, as supply is already constrained during the shift between regions. Figure 1 illustrates this seasonal rotation. Salinas dominates shipments from roughly weeks 15–50, while Western Arizona dominates the winter weeks. Price spikes are concentrated around the transition periods, when aggregate supply is temporarily reduced.

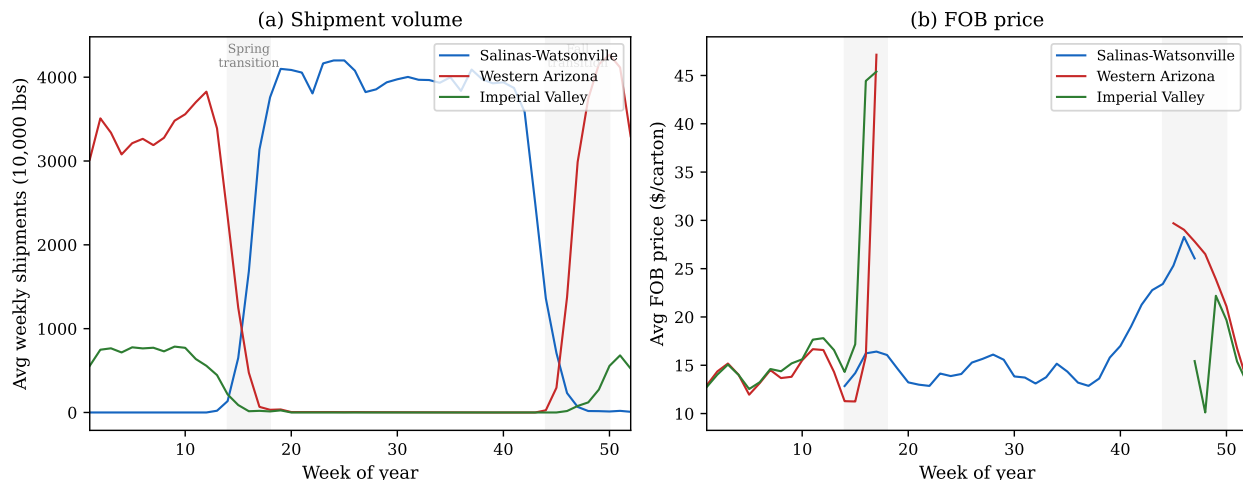


Figure 1: Seasonal shipment and price patterns by production district. Panel (a) shows average weekly shipment volume and panel (b) shows average FOB price, both by week of year across the 2010–2026 sample. Gray bands indicate the spring (weeks 14–18) and fall (weeks 44–50) transition periods.

Lettuce has a short growing cycle of approximately 60 to 90 days from planting to harvest.

Unlike annual grain crops, production occurs on a continuous, rolling basis throughout the season, with fields replanted multiple times per year. As a result, there is no single harvest event that determines total output. Instead, supply in any given week reflects planting decisions made two to three months earlier, realized growing conditions, and the pace of harvest operations. This structure generates a natural lag between weather shocks and realized supply.

This production environment has an important implication for empirical analysis. High-frequency yield data, which are available for many grain crops at the county level, do not exist for lettuce. There is no weekly or even monthly measure of production by district. Shipment data, which record the volume of lettuce loaded onto trucks at the point of origin, provide the closest available proxy for realized supply. Because lettuce is highly perishable, with a shelf life of roughly one to two weeks, storage is limited and most production is shipped shortly after harvest. Shipment volumes therefore provide a high-frequency measure of supply that is central to our empirical strategy.

4 Data

4.1 Sources

We combine three publicly available data sources covering January 2010 through April 2026.

Shipment volume. Weekly truck shipment reports from the USDA Agricultural Marketing Service (USDA Agricultural Marketing Service, 2026*b*) record the volume of each commodity shipped by truck from each production district. These reports provide the highest-frequency measure of supply available for lettuce. Unlike grain crops, which have annual county-level yield data, lettuce is harvested continuously within each growing season and no comparable high-frequency production data exist. Shipment volume therefore serves as a proxy for realized supply: because lettuce is highly perishable and cannot be stored for more than a few days, most production is shipped within the week it is harvested.

Prices. Daily FOB (free on board) shipping point prices from USDA Market News (USDA Agricultural Marketing Service, 2026a) report the price at which lettuce is sold at the point of origin, before transportation costs. We focus on FOB prices rather than retail prices because they more directly reflect production-side conditions. Prices are reported by district, commodity, and package type; we aggregate daily observations to weekly averages.

Weather. Daily temperature and precipitation data come from the PRISM Climate Group at Oregon State University (Daly et al., 2008), which provides gridded climate data at 4 km resolution for the contiguous United States. For each production district, we extract daily maximum temperature, minimum temperature, and precipitation at the district centroid and aggregate these variables to weekly measures. This construction aligns weather exposure with the timing of shipment and pricing outcomes.

4.2 Production Districts

The data cover six lettuce production districts in California and Arizona: Salinas-Watsonville, Western Arizona (Yuma), Imperial Valley, Santa Maria, Oxnard, and the San Joaquin Valley. Together, these districts account for the vast majority of U.S. lettuce production.

District activity and reporting vary substantially over time. Salinas-Watsonville, the largest district, ships in 560 of the 849 sample weeks and reports prices in 91% of those weeks. Western Arizona ships in 429 weeks, primarily during the winter months, with 82% price coverage. The remaining districts have more limited and irregular reporting: Oxnard reports prices in only 2% of its active weeks, Santa Maria in 69%, and the San Joaquin Valley in 76%.

This uneven coverage has two implications. First, price data are missing non-randomly across districts and time. Second, relying on any single district would substantially reduce the usable sample. These features motivate the dual-layer empirical strategy described in Section 5.1, which combines market-level aggregation to maximize coverage with district-level variation for identification.

4.3 Variable Construction

Shipment volume is measured in units of 10,000 pounds, following USDA refrigerated truck movement reports.

Market-level price. For each week, we construct a shipment-weighted average FOB price across all districts that report a price (Equation 1). Weighting by shipment volume ensures that districts with larger production have greater influence on the market price. To account for missing price observations, we define a coverage ratio (Equation 2) as the share of total weekly shipments for which a price is observed. The mean coverage ratio is 0.933, and 98.2% of weeks have at least one district reporting a price. This construction yields a consistent weekly price series while also measuring data quality through the coverage ratio.

Market-level weather. Weather variables are aggregated to the market level using shipment weights. Market-level weather reflects conditions in the districts that are actively supplying the market in a given week, rather than a fixed geographic average. When production shifts seasonally from Salinas to Yuma, the market-level weather measures shift accordingly.

Extreme weather indicators. Using the daily weather data, we construct three indicator variables for each district-week. The variable *freeze_risk* equals one if the minimum temperature falls below 0°C on any day within the week. The variable *extreme_heat* equals one if the maximum temperature exceeds 40°C on any day within the week. The variable *heavy_rain* equals one if total weekly precipitation exceeds 25 mm. At the market level, each indicator equals one if any active district experiences the event.

4.4 Summary Statistics

Table 1 reports summary statistics for the market-level dataset. The mean weekly iceberg lettuce FOB price is \$16.27 per carton with a standard deviation of \$11.80. The distribution is right-skewed, with a median of \$12.30 and a maximum of \$94.33, reached during the 2022 INSV outbreak. Mean weekly shipment volume is 4,422 (in 10,000 lb units). Extreme

weather events are relatively infrequent at the market level: freeze risk occurs in 14 of 849 weeks (1.6%), extreme heat in 6 weeks (0.7%), and heavy rain in 68 weeks (8.0%).

Table 1: Summary statistics: Market-level dataset (849 weekly observations, 2010–2026)

Variable	Mean	SD	Min	Max
Price (\$/carton)	16.27	11.80	5.99	94.33
Volume (10,000 lbs)	4,422	749	2,313	6,993
Coverage ratio	0.93	0.19	0.00	1.00
Tmax (°C)	21.90	3.20	14.80	35.70
Tmin (°C)	10.40	2.50	1.90	17.70
Precipitation (mm)	1.95	5.93	0.00	80.52
Active districts	2.90	1.10	2.00	6.00

5 Methodology and Empirical Strategy

This section describes the empirical framework. Section 5.1 outlines the dual-layer design that motivates the two datasets. Sections 5.2–5.4 present the model specifications. Section 5.5 describes feature construction, and Section 5.6 details the out-of-sample validation strategy.

5.1 Dual-Layer Design

The U.S. lettuce market spans six production districts across California and Arizona, but not all districts report FOB prices every week. In particular, the three smaller districts (Santa Maria, Oxnard, and San Joaquin Valley) exhibit substantial gaps in price reporting. A district-level panel including all six districts would have a price coverage rate of only 78.4%, making it unsuitable for time-series prediction.

We address this by constructing two datasets from the same underlying weekly panel. The first aggregates all six districts into a single market-level time series. The second retains the three core districts (Salinas-Watsonville, Western Arizona, and Imperial Valley) as a panel.

Dataset A: Market-level aggregate. Each week, we compute a shipment-weighted average price across all districts that report a price:

$$P_t = \frac{\sum_i V_{it} \cdot P_{it}}{\sum_i V_{it}}, \quad (1)$$

where V_{it} is iceberg lettuce shipment volume from district i in week t , and the sum runs over districts with nonmissing prices. Weather variables are aggregated analogously using shipment weights. This produces 849 weekly observations with 98.2% price coverage. The coverage ratio,

$$C_t = \frac{\sum_{i \in \text{priced}} V_{it}}{\sum_i V_{it}}, \quad (2)$$

measures the share of total shipments for which a price is observed and serves as a data quality indicator.

Dataset B: District-level panel. We retain the three districts with both substantial shipment volume and relatively consistent price reporting: Salinas-Watsonville, Western Arizona, and Imperial Valley. This panel contains 1,466 observations with 78.4% price coverage.

The two datasets serve distinct roles. Dataset A is used for prediction, where complete coverage and a continuous time series are essential. Dataset B is used for identification, where cross-district variation in weather provides the identifying variation.

5.2 Market-Level Prediction Model

The market-level price model takes the form

$$P_t = \alpha + \beta'_1 \mathbf{W}_t + \beta'_2 \mathbf{L}_t + \beta'_3 \mathbf{S}_t + \beta'_4 \mathbf{X}_t + \varepsilon_t, \quad (3)$$

where P_t is the shipment-weighted FOB price in week t , $\mathbf{W}_t$ is a vector of contemporaneous weather variables, $\mathbf{L}_t$ contains lagged price and volume terms, $\mathbf{S}_t$ includes seasonal indicators,

and $\mathbf{X}_t$ contains controls such as the coverage ratio and the number of active districts.

We estimate Equation (3) by OLS with heteroskedasticity-robust standard errors (White, 1980). We also estimate the same specification using XGBoost (Chen and Guestrin, 2016) and Random Forest (Breiman, 2001) as nonlinear benchmarks.

For the tree-based models, we use conservative hyperparameters to limit overfitting. The XGBoost model uses 200 trees, maximum depth 3, learning rate 0.03, subsample ratio 0.7, and L1/L2 regularization ($\alpha = 1.0$, $\lambda = 5.0$). The Random Forest uses 500 trees, maximum depth 6, and minimum leaf size 10. These settings are intentionally restrictive, as the goal is to evaluate out-of-sample performance rather than maximize in-sample fit.

5.3 Change Model

Lettuce prices are highly persistent: the first-order autocorrelation exceeds 0.9, and in level regressions the lagged price explains most of the variation. This persistence can mask the effects of other variables, including weather. To isolate the role of short-run shocks, we estimate models in first differences:

$$\Delta P_t = P_t - P_{t-1} = \alpha + \boldsymbol{\gamma}'\mathbf{W}_t + \boldsymbol{\delta}'\mathbf{M}_t + u_t, \quad (4)$$

where $\mathbf{W}_t$ contains weather variables and $\mathbf{M}_t$ includes volume, coverage, and seasonal controls. Lagged prices are excluded because differencing removes the autoregressive component.

Comparing the level and change specifications allows us to assess whether weather effects are attenuated by price persistence.

5.4 District-Level Panel

The district panel exploits cross-sectional variation in weather exposure that is averaged out at the market level. A freeze event in Yuma, for example, affects local prices directly but is diluted when aggregated with regions that do not experience the same shock.

Identification relies on the assumption that short-run weather variation affects supply but not demand. Weather shocks influence prices through crop damage and harvest disruptions rather than contemporaneous shifts in consumer demand. This assumption is standard in the agricultural economics literature and is plausible at the weekly frequency considered here.

The district panel specification is

$$\Delta P_{it} = \phi' \mathbf{W}_{it} + \alpha_i + \mu_q + \eta_{iq} + \nu_{it}, \quad (5)$$

where ΔP_{it} is the week-over-week price change for district i in week t , $\mathbf{W}_{it}$ is the local weather vector, α_i are district fixed effects, μ_q are quarter fixed effects, and η_{iq} captures district-by-season interactions. Standard errors are clustered at the district level. Identification comes from cross-district variation in weather within the same week.

We also estimate a level specification that includes lagged prices:

$$P_{it} = \phi' \mathbf{W}_{it} + \psi' \mathbf{L}_{it} + \alpha_i + \mu_q + \eta_{iq} + \nu_{it}. \quad (6)$$

Comparing the level and change specifications at the district level provides an additional check on whether persistence attenuates estimated weather effects.

5.5 Feature Construction

All models use the same set of constructed features.

Weather variables. Weekly weather includes maximum and minimum temperature ($tmax$, $tmin$), total precipitation (ppt), and diurnal range ($tmax - tmin$). Three indicators capture extreme conditions: *freeze_risk* equals one if minimum temperature falls below 0°C during the week, *extreme_heat* equals one if maximum temperature exceeds 40°C, and *heavy_rain* equals one if weekly precipitation exceeds 25 mm.

Lagged variables. For prices, shipment volume, temperature, and precipitation, we

include lags at 1, 2, and 4 weeks to capture short-run persistence and the production cycle.

Rolling averages. Four-week rolling means of volume and weather variables capture the recent background state of the market.

Market structure. The coverage ratio C_t and the number of active districts n_t enter as controls.

Seasonality. Month and week-of-year enter as continuous variables to capture the strong seasonal pattern in production and prices.

5.6 Validation Strategy

Out-of-sample performance is evaluated using an expanding-window approach. For each test year τ from 2014 to 2026, models are trained on all prior years and evaluated on year τ , mimicking a real-time forecasting setting.

We require a minimum of four years of training data, so the first test year is 2014 (trained on 2010–2013). The expanding window is preferred over a rolling window because the sample is relatively short and additional training data improves stability; we verify this in Section 8.

Model performance is measured using root mean squared error (RMSE) in price levels. For change models, predictions are converted back to price levels before evaluation. The naive baseline predicts $\hat{P}_t = P_{t-1}$, corresponding to zero predicted price change.

6 Results

6.1 Market-Level Price Prediction

Table 2 reports average out-of-sample prediction errors for the market-level price models over the 2014–2026 test period. The naive baseline, which predicts $\hat{P}_t = P_{t-1}$, has an average RMSE of \$5.11. The OLS level model with the full feature set achieves an RMSE of \$4.36, a 14.7% improvement over the naive baseline and the best performance among the models considered.

Both tree-based methods perform worse than OLS in level models. XGBoost averages \$6.67 and Random Forest \$6.81, each substantially worse than the naive baseline. This pattern is not driven by hyperparameter choice. The conservative settings we use are designed to limit overfitting, and the models capture meaningful in-sample variation. Rather, lettuce prices are driven primarily by a strong autoregressive component, and tree-based models do not perform well in this setting out of sample.

The year-by-year results reveal the source of the gap most clearly in 2022. That year, an outbreak of Impatiens Necrotic Spot Virus (INSV) in the Salinas Valley disrupted production severely, and the weekly FOB price for iceberg lettuce reached \$94.33, roughly five times the sample mean. OLS tracked the spike with an RMSE of \$7.77 for that year, while XGBoost and Random Forest produced much larger errors, with RMSEs of \$24.37 and \$23.83, respectively. Because the training data contain no comparable price spike, the tree-based models substantially underpredict the 2022 peak. This illustrates a limitation of tree-based methods in small-sample time series settings: they are generally poor at extrapolating beyond the range of the training data, while linear autoregressive models can extend trends through the lagged price coefficient.

Table 2: Out-of-sample model comparison: average RMSE and MAE (2014–2026, expanding window)

Model	Avg RMSE (\$)	Avg MAE (\$)	Improvement vs. Naive
Naive ($P_t = P_{t-1}$)	5.11	3.35	baseline
<i>Level models</i>			
OLS (full)	4.36	3.14	+14.7%
XGBoost	6.67	4.26	−30.5%
Random Forest	6.81	4.32	−33.3%
<i>Change models</i>			
OLS ΔP	5.07	3.57	+0.8%
XGBoost ΔP	5.12	3.56	−0.2%
Random Forest ΔP	5.00	3.40	+2.2%
OLS log ΔP	5.00	3.41	+2.2%

Figure 2 plots actual and predicted market-level prices from the OLS model. The model

tracks normal years closely, while most large errors occur during the 2022 price spike driven by the INSV outbreak.

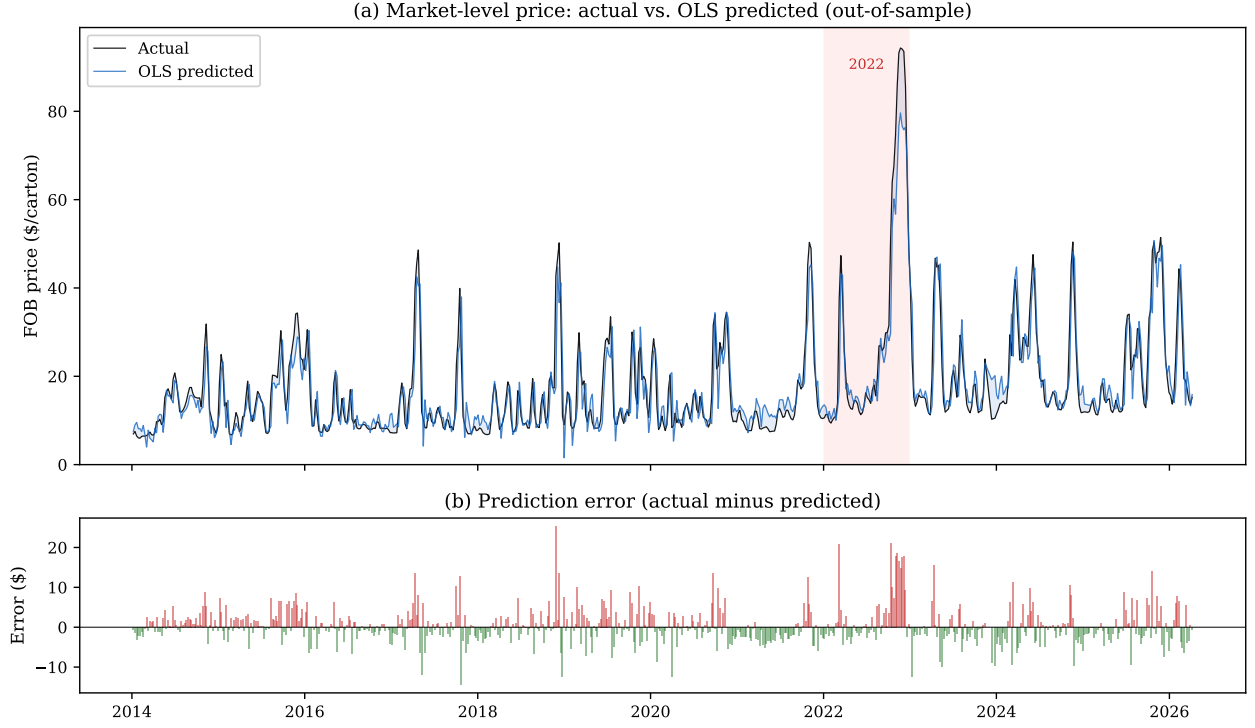


Figure 2: Market-level iceberg lettuce price: actual vs. OLS predicted (out-of-sample, expanding window, 2014–2026). Panel (a) shows the price time series; the red band highlights 2022. Panel (b) shows prediction errors (actual minus predicted).

6.2 Price Persistence and Hidden Weather Effects

In the full level regression, the lagged price coefficient is 1.369 ($p < 0.001$), and the in-sample R^2 is 0.891. The one-week lagged price alone accounts for 30% of total feature importance in the XGBoost level model, and the three price lags together account for 45%, indicating that price persistence dominates the predictive structure. Precipitation is statistically significant ($\hat{\beta} = 0.123$, $p = 0.010$), but its economic magnitude is small relative to the lagged price term.

The change models tell a different story. When we difference out price persistence by modeling ΔP_t instead of P_t , two things happen. First, all methods converge: OLS, XGBoost,

Random Forest, and the log-differenced OLS all produce average RMSEs within a narrow range of \$5.00 to \$5.12. The gap between OLS and tree-based methods disappears because the dominant linear feature, lagged price, has been removed. The residual relationship between weather and price changes is approximately linear, so nonlinear methods offer no advantage.

Second, the relative importance of weather variables increases substantially. In the level model, temperature ($tmax$) contributes only 2.8% of feature importance. In the change model, where price lags are excluded by construction, temperature importance rises to 6.0%, and other weather variables (precipitation, diurnal range) increase as well. This confirms that the weather signal is present in the data but is masked by price persistence in the level specification. Figure 3 visualizes this shift.

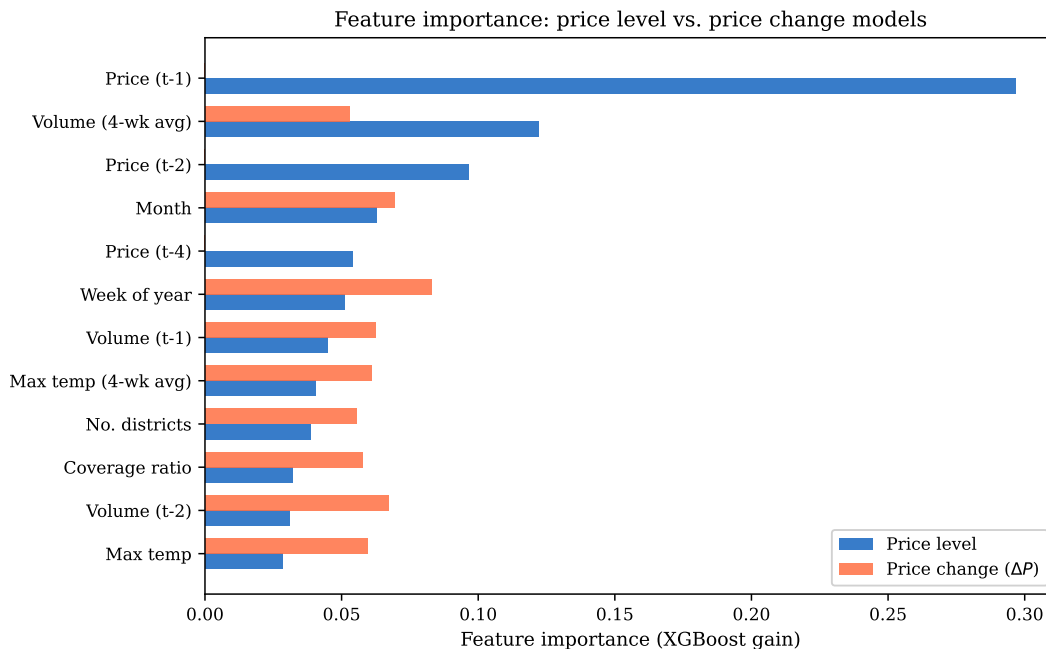


Figure 3: XGBoost feature importance: price level vs. price change models. The level model is dominated by lagged prices, particularly Price ($t-1$). In the change model, where price lags are removed by construction, importance is redistributed toward weather, volume, and seasonal variables.

6.3 Volume Prediction

Table 3 reports results for market-level shipment volume prediction. The pattern mirrors the price results: OLS outperforms XGBoost. The OLS level model achieves an average RMSE of 287, a 38.0% improvement over the naive baseline of 463. XGBoost achieves 354, better than the naive baseline but well behind OLS.

Precipitation is the most consistent weather predictor of volume. In the OLS level model, the coefficient on weekly precipitation is -9.0 (in 10,000 lb units) per millimeter ($p = 0.015$). Heavy rain disrupts field operations and harvest logistics, reducing the volume of lettuce that can be shipped in a given week. Freeze risk has a large negative point estimate (-195) but is not statistically significant at conventional levels ($p = 0.12$), likely because freeze events are infrequent in the sample.

Table 3: Volume prediction: average RMSE and MAE (10,000 lbs, expanding window, 2014–2026)

Model	Avg RMSE (10,000 lbs)	Avg MAE (10,000 lbs)	Improvement vs. Naive
Naive ($V_t = V_{t-1}$)	463	347	baseline
OLS Level	287	223	+38.0%
XGBoost Level	354	265	+23.6%
OLS ΔV	459	345	+0.8%
XGBoost ΔV	453	345	+2.1%

7 Mechanism: Weather, Supply, and Price

The results in Section 6 establish that weather matters for lettuce prices but that its effect is masked by price persistence in level models. This section uses the district-level panel to identify how weather operates, and to explain why the market-level aggregate understates weather effects.

7.1 District-Level Identification

Table 4 reports weather coefficients from the district panel specifications described in Section 5.4. The change model (Equation 5) includes district fixed effects, quarter fixed effects, and district-by-season interactions. Standard errors are clustered at the district level.

Freeze risk is the strongest and most robust weather predictor. In the change model, a week with at least one day below freezing is associated with a price increase of \$2.33 ($p < 0.001$). This estimate is stable across specifications: adding volume controls and lagged weather changes the coefficient only slightly, from \$2.33 to \$2.37 to \$2.24. Minimum temperature has a significant negative coefficient ($-\$0.15$ to $-\$0.17$ per degree Celsius, $p < 0.01$), which is the continuous counterpart of the freeze risk indicator: colder weeks are associated with higher prices. Extreme heat is significant in the level specification (\$2.16, $p = 0.008$) but not in the change models, suggesting that heat effects operate over a slightly longer horizon than week-to-week changes can capture.

Table 4: District-level weather coefficients (Iceberg, OLS with clustered SE)

Variable	ΔP_{it} (M1)	$\Delta P_{it} + \text{vol}$ (M2)	+ weather lags (M3)	Level (M4)
freeze_risk	2.326***	2.368***	2.237***	1.354
tmin_avg	-0.164**	-0.169***	-0.145***	-0.128***
extreme_heat	0.542	0.493	0.778	2.164***
ppt_total	0.106	0.105	0.104	0.089
heavy_rain	-0.914	-0.919	-0.824	-2.124

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All models include district FE, quarter FE, and district $\times$ season interactions. SE clustered by district.

The precipitation coefficient is positive but not statistically significant in any specification. This result is somewhat unexpected given that precipitation significantly predicts both price at the market level and volume. A plausible explanation is that precipitation effects on price operate primarily through volume reductions at the market level, and the district panel, which does not aggregate across districts, captures a noisier version of this channel.

7.2 Why Level Models Mask Weather Effects

The comparison across columns in Table 4 illustrates how price persistence attenuates weather coefficients. Freeze risk is large and highly significant in all three change specifications but shrinks and loses significance in the level model (column M4). The level model includes lagged prices, which absorb much of the variation that weather explains in the change model. Since a freeze event raises prices this week, and lagged price carries that increase forward into next week, the level model attributes the persistent effect to the lag rather than to the freeze.

This attenuation is not a statistical artifact. It reflects a real feature of the price process. Weather shocks are transitory, but their price effects persist through the autoregressive structure of the market. A freeze in Yuma raises the Yuma price, which feeds into the market-level price, which then carries forward through price persistence. The level model captures the persistence but not the initial cause. The change model captures the cause but not the persistence. Both perspectives are necessary for a complete picture of the price process.

The district panel also reveals heterogeneity that is lost in the market aggregate. A freeze event in Western Arizona has a direct and immediate effect on the local price, but when averaged with Salinas (which may not experience a freeze in the same week), the signal is diluted. This helps explain why freeze risk is highly significant ($p < 0.001$) in the district panel but not significant in the market-level regressions. The dual-layer design is not just a convenience for handling missing data; it is necessary for identifying localized weather effects that aggregation would obscure.

Taken together, the evidence supports the following mechanism. Weather shocks, particularly freezes and heat waves, reduce local supply in the affected district, leading to increases in local prices. Because the market is spatially connected (buyers can substitute across districts to some extent), the price increase propagates to the market level and persists through the autoregressive price process. This mechanism reconciles the apparent disconnect between

weak weather effects in aggregate regressions and strong effects in the panel estimates.

8 Robustness

8.1 Romaine Lettuce

We replicate the core analysis using Romaine lettuce prices instead of Iceberg. Table 5 compares the two varieties. The market-level OLS model achieves an out-of-sample RMSE of \$2.61 for Romaine, compared to \$4.36 for Iceberg. The lower RMSE reflects lower price volatility rather than improved model fit; the in-sample R^2 is actually higher for Romaine (0.952 vs. 0.891).

Table 5: Romaine vs. Iceberg: model performance and weather coefficients

	Iceberg	Romaine
<i>Model performance</i>		
In-sample R^2 (level)	0.891	0.952
In-sample RMSE	\$3.97	\$2.48
OOS avg RMSE	\$4.36	\$2.61
Price mean	\$16.27	\$17.80
Price coverage	98.2%	98.4%
<i>Key coefficients</i>		
price_lag1 (level)	1.369***	1.633***
ppt_total (ΔP)	0.192***	0.137***
freeze_risk (ΔP)	1.176	1.843**
heavy_rain (ΔP)	-2.003**	-1.647***

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The key finding is that the weather coefficients have consistent signs across the two varieties. Precipitation is positive and significant for both ($p < 0.01$), freeze risk is positive in both (and significant for Romaine), and heavy rain is negative and significant in both. The core findings of this paper, that weather affects prices through supply disruptions and that these effects are masked by price persistence, are not specific to Iceberg lettuce. They reflect the structure of the lettuce market more broadly.

8.2 Expanding vs. Rolling Window

Table 6 compares expanding-window and five-year rolling-window validation. The expanding window produces a lower average RMSE (\$4.36 vs. \$4.57). The two approaches yield nearly identical results in early test years, when they share the same training set. The gap widens in later years: in 2021, the expanding window achieves \$3.41 vs. \$4.61 for rolling, and in 2022 the gap is \$7.77 vs. \$9.22. The expanding window benefits from a larger training sample, and the data-generating process appears sufficiently stable that older observations remain informative.

Table 6: Expanding vs. rolling window: year-by-year RMSE (\$)

Year	Expanding		Rolling (5yr)	
	RMSE	MAE	RMSE	MAE
2014	2.47	1.86	2.47	1.86
2015	3.60	2.83	3.60	2.83
2016	2.66	1.84	2.71	1.85
2017	4.92	3.40	5.00	3.48
2018	5.12	2.94	5.00	2.88
2019	4.61	3.71	4.69	3.78
2020	4.17	2.95	4.27	2.97
2021	3.41	2.71	4.61	3.92
2022	7.77	4.93	9.22	5.81
2023	4.62	3.16	4.67	3.15
2024	4.56	3.53	4.71	3.54
2025	4.16	2.97	4.12	2.88
2026	4.59	4.04	4.28	3.77
Average	4.36	3.14	4.57	3.29

8.3 Coverage and Prediction Error

Table 7 shows that weeks with low price coverage are harder to predict. When the coverage ratio falls below 0.5, the average MAE is \$4.98, nearly double the \$2.83 for weeks with coverage above 0.95. This pattern is consistent with measurement error in the shipment-weighted price: when fewer districts report prices, the weighted average is noisier, and the

model has less reliable training signal. Including the coverage ratio as a feature helps the model partially adjust for this, but it cannot fully compensate for missing price information.

The low-coverage bin (0.5–0.7) contains only one observation and should be interpreted with caution. Figure 4 shows the underlying scatter, confirming the monotonic relationship between coverage and prediction accuracy.

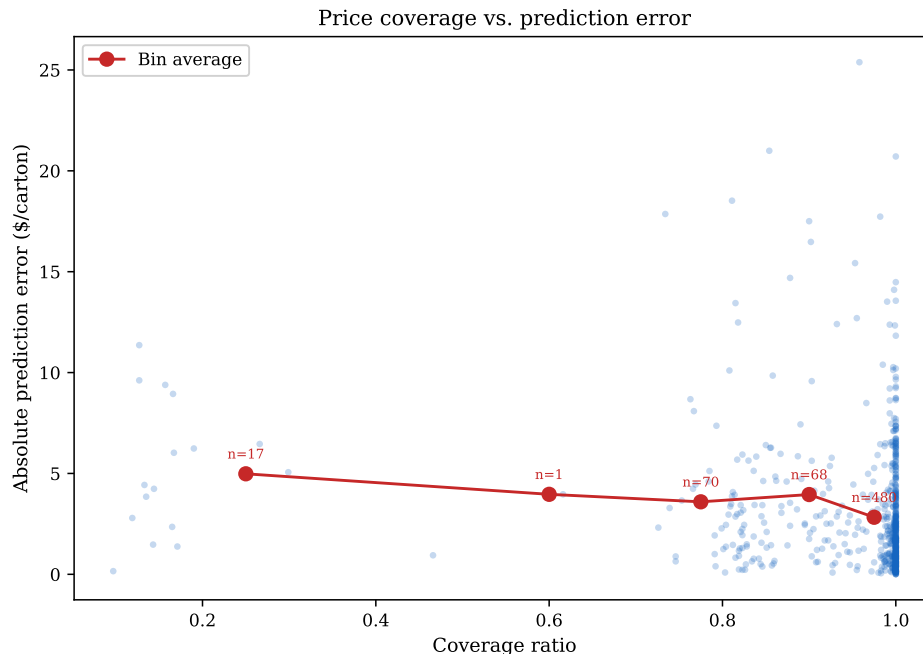


Figure 4: Price coverage ratio vs. absolute prediction error. Each point is one out-of-sample week. Red markers show bin averages with sample sizes. Prediction errors decline monotonically as coverage increases, consistent with measurement error in the shipment-weighted price.

Table 7: Prediction error by coverage ratio

Coverage bin	N	MAE (\$)	Median error (\$)	MAPE (%)
Very low (< 0.5)	17	4.98	4.43	29.6
Low (0.5–0.7)	1	3.97	3.97	16.8
Medium (0.7–0.85)	69	3.55	2.56	18.2
High (0.85–0.95)	69	3.99	2.87	19.5
Very high (> 0.95)	480	2.83	1.84	18.1

8.4 The 2022 Extreme Year

The 2022 crop year was an outlier by any measure. Table 8 summarizes the key statistics. Weekly iceberg lettuce prices averaged \$31.89 with a standard deviation of \$27.09, compared to typical years where the mean is around \$15 and the standard deviation is around \$5. The maximum weekly price reached \$94.33, driven by an outbreak of Impatiens Necrotic Spot Virus (INSV) in the Salinas Valley that severely disrupted production.

Table 8: 2022 extreme year: model performance comparison

	2022	Other years
<i>Price statistics</i>		
Mean price	\$31.89	~\$15
Max price	\$94.33	~\$50
Price std	\$27.09	~\$5
<i>Model RMSE</i>		
OLS Level	\$7.77	\$3.95
XGBoost Level	\$24.37	\$4.75
Random Forest Level	\$23.83	\$4.89
<i>Prediction error (OLS)</i>		
MAE	\$4.93	\$2.93

The disparity between OLS and tree-based methods in 2022 highlights an extrapolation issue: tree-based models are generally unable to extrapolate beyond the range of their training data, while OLS can follow the upward movement through the autoregressive component. We do not interpret 2022 as a failure of the model. Extreme phytosanitary shocks are inherently difficult to forecast using weather and market data alone. The relevant finding is that OLS performs better under extreme conditions, rather than producing highly unstable predictions.

8.5 Feature Stability

We split the sample into three sub-periods and re-estimate the market-level OLS model on each. Table 9 reports the key coefficients. The lagged price coefficient ranges from 1.153 to

1.517 across periods, and the precipitation coefficient from 0.116 to 0.181. There is no clear evidence of a structural break in the core relationships.

The temperature variables ($tmax$, $tmin$) show sign instability, which likely reflects collinearity between them rather than instability in the underlying relationship. Freeze risk also varies across periods, consistent with its low frequency making it sensitive to individual events within each sub-sample.

Table 9: Coefficient stability across sub-periods (market-level OLS)

Variable	2010–2015	2016–2020	2021–2026
price_lag1	1.218	1.153	1.517
ppt_total	0.116	0.118	0.181
freeze_risk	1.682	−1.793	−1.355
heavy_rain	−2.190	−0.670	−2.181
extreme_heat	−0.000	−0.000	−0.160

8.6 Missing Price Prediction

Not all district-week observations have reported prices. Out of 2,441 district-week observations with positive shipment volume, 682 (28%) are missing a price. We train a model on observed prices using weather, volume, district indicators, and seasonal controls, then predict prices for the missing observations. OLS achieves an average out-of-sample RMSE of \$11.08 (expanding-window validation), compared to \$13.27 for XGBoost.

Table 10 compares predicted and observed mean prices by district. The predicted prices for missing observations are systematically higher than the observed means. For example, the predicted mean for Oxnard’s missing weeks is \$17.72 vs. an observed mean of \$15.30. This pattern is consistent with non-random missingness in price reporting: districts may be less likely to report prices during periods of high prices or supply stress.

Table 10: Missing price prediction by district

District	Missing obs	Predicted mean (\$)	Observed mean (\$)
Oxnard	238	17.72	15.30
Santa Maria	199	19.57	13.86
Western Arizona	76	18.93	17.26
Imperial Valley	71	20.93	15.49
Salinas-Watsonville	49	23.72	16.14
San Joaquin Valley	49	22.18	20.40
Total	682		

OLS RMSE = \$11.08, XGBoost RMSE = \$13.27 (expanding window).

9 Conclusion

This paper studies weather-driven supply responses in a setting where high-frequency yield data are not available. In contrast to major row crops such as corn and soybeans, lettuce production is geographically concentrated and occurs as a continuous flow throughout the season rather than as a single annual harvest. As a result, standard approaches that rely on county-level yield data are not feasible. Instead, we use weekly shipment volumes and FOB prices as downstream market outcomes to infer supply responses. Identification relies on the assumption that short-run weather variation affects supply but not demand. Under this assumption, variation in prices and shipments can be used to recover the impact of weather on supply.

We document three main results. First, simple linear models outperform tree-based machine learning methods for price prediction. OLS achieves an out-of-sample RMSE of \$4.36, compared to \$6.67 for XGBoost and \$6.81 for Random Forest. Lettuce prices exhibit strong persistence, and most of the predictable variation is captured by lagged prices. Conditional on this persistence, the remaining relationship between weather and prices is approximately linear, so more flexible methods do not improve predictive performance and perform poorly during extreme events.

Second, weather effects on prices are real but masked by price persistence. In level models, the three price lags together account for nearly half of total feature importance, leaving

limited scope for weather variables to contribute. When we model price changes instead of price levels, weather variables become substantially more important. At the district level, freeze risk increases weekly prices by \$2.33 ($p < 0.001$), but this effect is attenuated when aggregated to the market level. The dual-layer approach, combining market-level aggregation for prediction with district-level panels for identification, is therefore necessary to uncover localized weather effects.

Third, these findings are robust. The results replicate for Romaine lettuce, coefficients are stable across sub-periods, and alternative validation strategies yield similar conclusions. Data quality, as measured by price coverage, is also an important determinant of prediction accuracy.

Several limitations remain. The identification assumption that weather affects supply but not demand is plausible at the weekly frequency considered here, but longer-run climate trends may affect both sides of the market. Our estimates are based on short-run weather variation and therefore do not capture long-run climate adaptation. They provide a useful benchmark for understanding climate-related supply risks, but extrapolating from weather responses to climate responses remains an open question. The 2022 virus outbreak highlights that non-weather shocks can dominate market outcomes, and such events are inherently difficult to predict using historical price and weather data. In addition, the analysis focuses on six production districts in California and Arizona; extending the framework to other crops or regions would require additional data and may yield different dynamics.

Despite these limitations, the results show that downstream market outcomes, shipment volumes and FOB prices, can be used to recover weather-driven supply responses even in the absence of high-frequency yield data. More broadly, the paper illustrates how market data can substitute for direct production measures in perishable specialty crops. For participants in fresh produce markets, the combination of publicly available USDA data and simple linear models provides a transparent and practical approach to monitoring weather-related price risk.

References

- Breiman, Leo.** 2001. “Random Forests.” *Machine Learning*, 45(1): 5–32.
- Chen, Tianqi, and Carlos Guestrin.** 2016. “XGBoost: A Scalable Tree Boosting System.” 785–794.
- Crane-Droesch, Andrew.** 2018. “Machine Learning Methods for Crop Yield Prediction and Climate Change Impact Assessment in Agriculture.” *Environmental Research Letters*, 13(11): 114003.
- Daly, Christopher, Michael Halbleib, Joseph I. Smith, Wayne P. Gibson, Matthew K. Doggett, George H. Taylor, Jan Curtis, and Phillip P. Pasteris.** 2008. “Physiographically Sensitive Mapping of Climatological Temperature and Precipitation across the Conterminous United States.” *International Journal of Climatology*, 28(15): 2031–2064.
- Kistner, E., O. Kellner, J. Andresen, D. Todey, and L. W. Morton.** 2018. “Vulnerability of Specialty Crops to Short-Term Climatic Variability and Adaptation Strategies in the Midwestern USA.” *Climatic Change*, 146(1): 145–158.
- Letta, Marco, Pierluigi Montalbano, and Guillaume Pierre.** 2022. “Weather Shocks, Traders’ Expectations, and Food Prices.” *American Journal of Agricultural Economics*, 104(3): 1100–1119.
- Lobell, David B., Kimberly Nicholas Cahill, and Christopher B. Field.** 2007. “Historical Effects of Temperature and Precipitation on California Crop Yields.” *Climatic Change*, 81(2): 187–203.
- Lobell, David B., Wolfram Schlenker, and Justin Costa-Roberts.** 2011. “Climate Trends and Global Crop Production since 1980.” *Science*, 333(6042): 616–620.

- Mullainathan, Sendhil, and Jann Spiess.** 2017. “Machine Learning: An Applied Econometric Approach.” *Journal of Economic Perspectives*, 31(2): 87–106.
- Ortiz-Bobea, Ariel, Toby R. Ault, Carlos M. Carrillo, Robert G. Chambers, and David B. Lobell.** 2021. “Anthropogenic Climate Change Has Slowed Global Agricultural Productivity Growth.” *Nature Climate Change*, 11: 306–312.
- Roberts, Michael J., and Wolfram Schlenker.** 2009. “World Supply and Demand of Food Commodity Calories.” *American Journal of Agricultural Economics*, 91(5): 1235–1242.
- Roberts, Michael J., and Wolfram Schlenker.** 2013. “Identifying Supply and Demand Elasticities of Agricultural Commodities: Implications for the US Ethanol Mandate.” *American Economic Review*, 103(6): 2265–2295.
- Roberts, Michael J., Wolfram Schlenker, and Jonathan Eyer.** 2013. “Agronomic Weather Measures in Econometric Models of Crop Yield with Implications for Climate Change.” *American Journal of Agricultural Economics*, 95(2): 236–243.
- Roll, Richard.** 1984. “Orange Juice and Weather.” *American Economic Review*, 74(5): 861–880.
- Schlenker, Wolfram, and Michael J. Roberts.** 2009. “Nonlinear Temperature Effects Indicate Severe Damages to U.S. Crop Yields under Climate Change.” *Proceedings of the National Academy of Sciences*, 106(37): 15594–15598.
- Storm, Hugo, Kathy Baylis, and Thomas Heckelei.** 2020. “Machine Learning in Agricultural and Applied Economics.” *European Review of Agricultural Economics*, 47(3): 849–892.

USDA Agricultural Marketing Service. 2026*a*. “Market News: Fruit and Vegetable Shipping Point Prices.” <https://marketnews.usda.gov/mnp/fv-home>, Accessed April 2026.

USDA Agricultural Marketing Service. 2026*b*. “Truck Rate Reports: Fruit and Vegetable Shipments.” <https://usda.library.cornell.edu/concern/publications/mg74qm08h>, Accessed April 2026.

USDA National Agricultural Statistics Service. 2023. “Vegetables Summary.” https://www.nass.usda.gov/Surveys/Guide_to_NASS_Surveys/Vegetables/, Accessed April 2026.

White, Halbert. 1980. “A Heteroskedasticity-Consistent Covariance Matrix Estimator and a Direct Test for Heteroskedasticity.” *Econometrica*, 48(4): 817–838.

Appendix

This appendix provides supplementary materials. Appendix A defines all variables used in the analysis. Appendix B reports the complete OLS coefficient estimates for the market-level price model. Appendix C presents out-of-sample RMSE by year for the six main models.

A Variable Definitions

Table 11: Variable definitions and data sources

Variable	Definition	Unit	Source
<i>Outcome variables</i>			
price	Shipment-weighted FOB price	\$/carton	USDA Market News
dprice	$P_t - P_{t-1}$	\$/carton	Constructed
volume	Total weekly truck shipments	10,000 lbs	USDA AMS
<i>Weather variables</i>			
tmax_avg	Weekly mean of daily max temperature	°C	PRISM
tmin_avg	Weekly mean of daily min temperature	°C	PRISM
ppt_total	Total weekly precipitation	mm	PRISM
diurnal_range	tmax_avg - tmin_avg	°C	Constructed
freeze_risk	$\mathbf{1}[\text{min temp} < 0^\circ\text{C any day}]$	Binary	Constructed
extreme_heat	$\mathbf{1}[\text{max temp} > 40^\circ\text{C any day}]$	Binary	Constructed
heavy_rain	$\mathbf{1}[\text{weekly precip} > 25 \text{ mm}]$	Binary	Constructed
<i>Lagged variables</i>			
price_lagk	$P_{t-k}, k \in \{1, 2, 4\}$	\$/carton	Constructed
volume_lagk	$V_{t-k}, k \in \{1, 2\}$	10,000 lbs	Constructed
tmax_lagk	tmax_avg _{t-k} , $k \in \{1, 2\}$	°C	Constructed
ppt_lag1	ppt_total _{t-1}	mm	Constructed
<i>Rolling averages</i>			
volume_roll4_mean	4-week rolling mean of volume	10,000 lbs	Constructed
tmax_avg_roll4_mean	4-week rolling mean of tmax	°C	Constructed
ppt_total_roll4_mean	4-week rolling mean of precip	mm	Constructed
<i>Market and seasonal</i>			
coverage	Share of shipments with observed price	[0, 1]	Constructed
n_districts	Number of active shipping districts	Count	Constructed
month	Calendar month	1–12	Constructed
week_of_year	Calendar week	1–52	Constructed

B Full OLS Regression Output

Table 12: Full OLS regression: market-level price model ($N = 823$, $R^2 = 0.891$)

Variable	Coeff.	Robust SE	z	p -value
<i>Price and volume lags</i>				
price.lag1	1.369	0.042	32.15	<0.001
price.lag2	-0.568	0.049	-11.64	<0.001
price.lag4	0.066	0.024	2.75	0.006
volume.lag1	-0.001	0.001	-1.33	0.185
volume.lag2	0.002	0.000	3.72	<0.001
<i>Contemporaneous weather</i>				
tmax_avg	21.197	28.093	0.75	0.451
tmin_avg	-21.389	28.103	-0.76	0.447
ppt_total	0.123	0.048	2.57	0.010
diurnal_range	-21.350	28.104	-0.76	0.447
extreme_heat	-0.580	0.784	-0.74	0.460
freeze_risk	0.836	1.304	0.64	0.521
heavy_rain	-1.478	0.688	-2.15	0.032
<i>Lagged weather</i>				
tmax.lag1	-0.275	0.097	-2.83	0.005
tmax.lag2	-0.106	0.110	-0.96	0.336
ppt.lag1	-0.017	0.033	-0.51	0.608
<i>Rolling averages</i>				
volume_roll4_mean	-0.002	0.001	-2.56	0.011
tmax_avg_roll4_mean	0.555	0.269	2.07	0.039
ppt_total_roll4_mean	-0.111	0.099	-1.11	0.266
<i>Market and seasonal</i>				
coverage	-1.406	1.417	-0.99	0.321
n_districts	0.233	0.188	1.24	0.215
month	0.113	0.137	0.82	0.410
week_of_year	-0.011	0.033	-0.33	0.741
constant	7.140	2.051	3.48	<0.001
RMSE = \$3.96. Dependent variable: shipment-weighted FOB price (\$/carton).				

C Out-of-Sample RMSE by Year

Table 13: Out-of-sample RMSE by year (\$, expanding window)

Year	Level models (full)			Change models (ΔP)		
	OLS	XGBoost	RF	OLS	XGBoost	RF
2014	2.47	2.74	2.78	2.76	2.80	2.71
2015	3.60	5.63	5.41	3.97	3.97	3.94
2016	2.66	2.79	2.92	3.21	3.32	3.12
2017	4.92	6.18	6.62	5.96	6.06	5.99
2018	5.12	6.20	6.52	6.25	6.16	6.21
2019	4.61	4.83	4.87	5.20	4.89	4.97
2020	4.17	4.87	4.93	4.78	4.43	4.49
2021	3.41	4.15	4.81	3.68	3.33	3.37
2022	7.77	24.37	23.83	6.96	7.25	7.30
2023	4.62	6.86	7.29	5.92	6.25	5.90
2024	4.56	6.57	6.43	5.72	5.89	5.67
2025	4.16	5.22	5.92	4.81	5.33	5.07
2026	4.59	6.35	6.23	6.69	6.84	6.23
Average	4.36	6.67	6.81	5.07	5.12	5.00